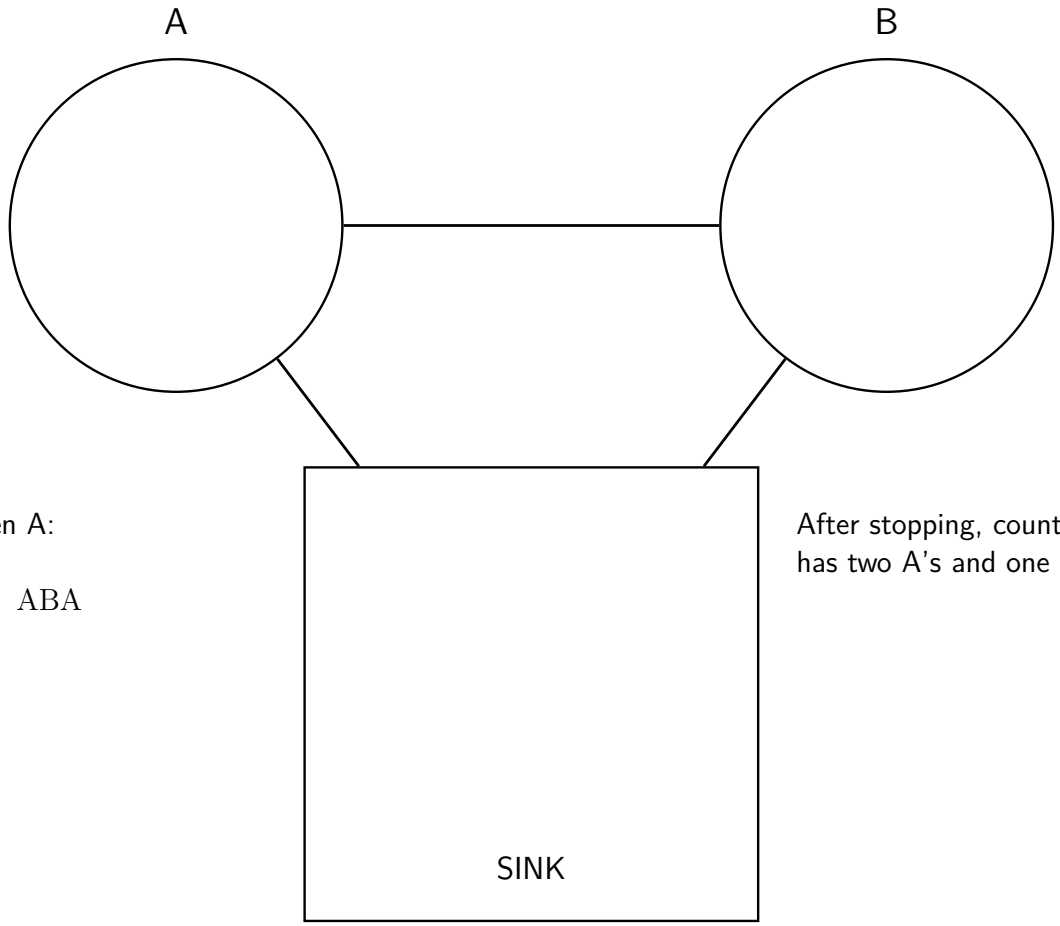
A circle needs one chip for every line touching it before it can share. To share, send one chip along each line. The sink keeps its chips and never shares. Choose one circle at a time. Stop when no circle can share. Empty the sink before each new start. Write each sharing letter in order, then the final piles.

Problem 1: Finish each start in two different orders, if possible. After stopping, count each letter in the sharing word. Can a different order change the final piles or these counts?



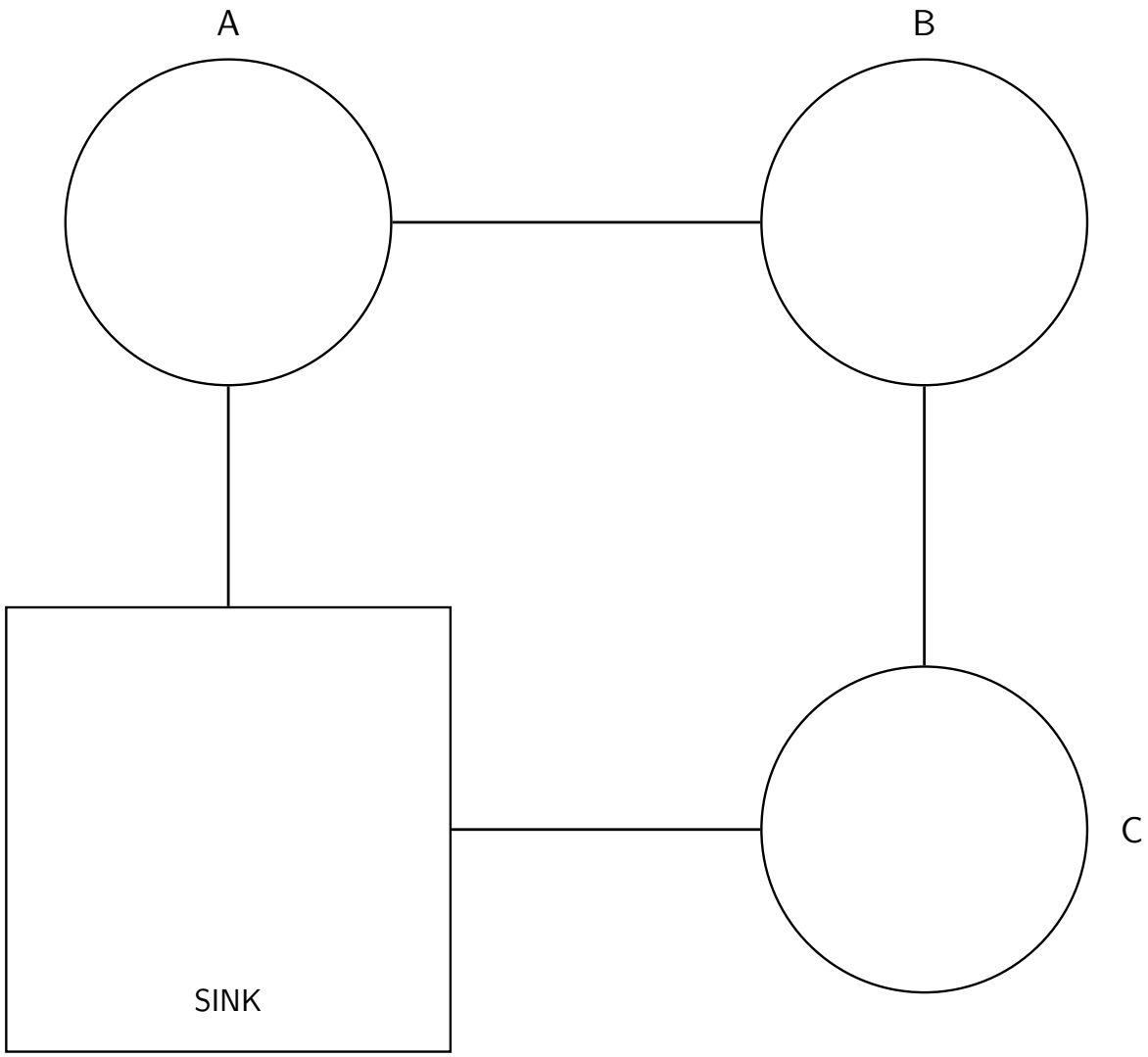
A, then B, then A:

$\mathrm{A} \rightarrow \mathrm{AB} \rightarrow \mathrm{ABA}$

After stopping, count letters: ABA has two A's and one B.

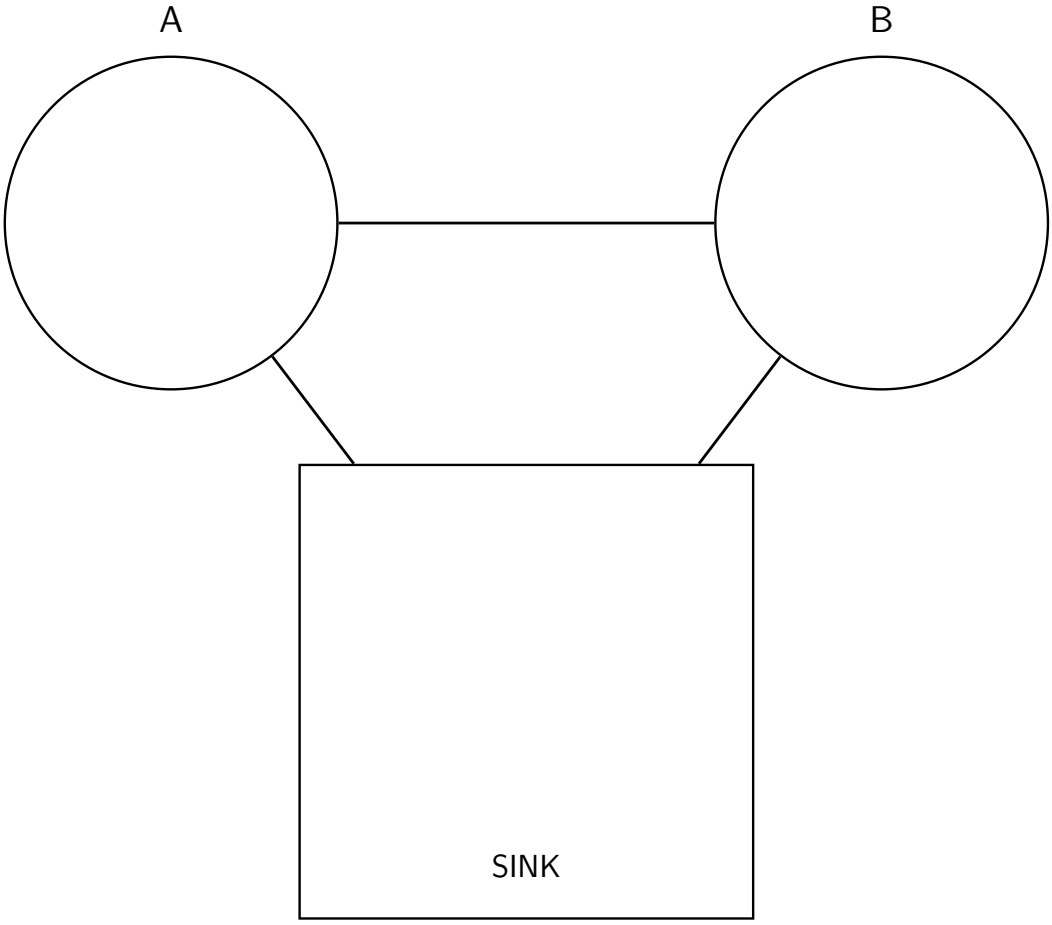
Start A	Start B	Sharing word	Finish A	Finish B
2	2			
2	2			
4	0			
4	0			
3	3			
3	3			

Problem 2: Finish each start in two different orders. After stopping, count each letter in the sharing word. Can a different order change the final piles or these counts?



Start A	Start B	Start C	Sharing word	Finish A	Finish B	Finish C
0	4	0				
0	4	0				
2	1	2				
2	1	2				
2	4	2				
2	4	2				

Problem 3: Put exactly 6 chips altogether on A and B. Find every start that finishes with one chip at A and one at B. Explain how you know you have found every start that works.



Start A	Start B	Finish A	Finish B